GECKO

Version 1.5.0 — Detailed Release Notes

Main Menu Updates

Session Resume Improvements

- **Resume prompt now shows trial number and display name** (e.g. '3. 07_02') instead of the raw internal trial identifier, matching how trials appear in the trial list.
- **Split resume prompt:** Resuming a session now asks two separate questions — whether to jump back to the previous trial, and whether to restore the previous 'Channels in Trial' inspection selections. This allows returning to a trial without overwriting current channel selections.
- **Session state is now saved on close** rather than on every channel or export click. This eliminates redundant disk writes during normal use while still preserving resume state.
- **Removed overlap between session_state.json and user_prefs.json.** Export channel and event marker defaults are now exclusively managed by user_prefs.json. Session state is now strictly a 'resume where you left off' record: it stores the last selected trial, the channel search filter text, and the inspect channel selections only.

  Note: If you have an existing session_state.json from a previous version, the export and marker fields will be ignored on load. Your selections from user_prefs.json will apply instead.

Per-Gap Interpolation Selection

- **Each large data gap is now handled individually.** Previously, one interpolation strategy was applied to all gaps in a trial. Users are now prompted separately for each gap (displayed as 'Gap X of Y: frames A-B'), allowing different strategies per gap within the same trial.

Gaze Labeler Channel Picker

- **Additional overlay channels can now be selected** for display in the gaze labeler. Gaze_X and Gaze_Y are always included. Users can select additional channels (e.g. xT, yT) from a dialog on first launch. This selection is persistent and can be cleared via Clear Cache.

Gaze Labeler Safety Dialog

- **'Restart All' now requires confirmation.** A confirmation dialog was added to the Restart All button in the gaze labeler to prevent accidental loss of labeled segments.

UI Performance Improvements

- **Interpolation window no longer flashes or reloads between gaps.** The preview figure is now created once and reused across all gaps in a channel, rather than being destroyed and recreated for each gap. The window stays open and maximized while you step through gaps.
- **Bug fix: each gap's strategy is now applied only to that gap.** Previously, selecting 'Linear' for gap 1 would apply linear interpolation to all gaps, then selecting a strategy for gap 2 would apply it to all gaps again — overwriting earlier decisions. Each gap's chosen strategy is now applied independently.

- **Help window now opens at the correct size on all screen resolutions.** The window previously opened compressed on smaller or high-DPI displays, requiring manual resizing to reach the Refresh button. A minimum size and screen-relative height are now enforced.

Exported Metrics Now Low-Pass Filtered

- **Computed channels (Angular Velocity, Spherical Coordinates, FVR) are now low-pass filtered in the export path** using the same 20 Hz Butterworth filter applied during interactive display. Previously there was a discrepancy between what was shown in the GUI and what was written to the exported CSV.

Backend & Code Quality (Internal)

These changes do not affect analysis outputs. They improve maintainability, configurability, and long-term reliability of the codebase.

Codebase Restructuring

- **gui_main.py split into focused modules.** The main GUI file (~1729 lines) has been decomposed into panel classes following a consistent composition pattern:
  - **gui_utils.py** — window centering, treeview styling, crash logging
  - **gui_trial_panel.py** — trial marking, notes, trial list refresh
  - **gui_channel_panel.py** — channel inspection, filtering, derived channel computation, plotting
  - **gui_export_panel.py** — export channel list, event marker list, action buttons
- gui_main.py is now ~750 lines and serves purely as an orchestration layer.

Bug Fixes

- **Butterworth filter order bug fixed.** The filter order was not being halved before passing to scipy's butter(), causing filtfilt to apply double the intended order (effective 8th-order instead of 4th-order). This affected all low-pass filtered channels.
- **'Clear Interpolation Cache' confirmation dialog** was not appearing due to a silent AttributeError (interpolation_methods vs interpolation_cache). Fixed.
- **Clear Cache dialog button layout fixed.** One button was being assigned None due to chained .pack() call, preventing subsequent buttons from rendering correctly.

Lab Configurability

- **Processing parameters centralized in user_prefs.py.** Constants such as filter cutoff frequencies, Savitzky-Golay parameters, foveal cone angle, and sentinel value thresholds are now defined in one place for lab-level adjustment without modifying core logic.
- **Per-arm conditional channel extraction** added to support single-arm lab configurations.

Dead Code & Import Cleanup

- Removed unused state variables: bad_trials, _current_file_id, _event_channel_map.
- Removed duplicate get_marker_defaults import and unused clear_interpolation_cache standalone method (superseded by clear_cache_dialog).
- Removed unused next_trial_name variable in label_gaze loop.

- Moved platform import to module level from inside __init__.
- Removed stale save_state calls on every listbox click event (export select, marker select, channel select). Session is now written only on trial change and application close.

DerivedChannel Dataclass

- DerivedChannel dataclass moved from gui_main.py to data_loader.py where it belongs, eliminating a GUI dependency on a data model definition.